

WASP-43b

R-2603

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140209 · created 2026-08-28T00:52:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456697.83097 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.027
Transit depth (Rp/Rs)^2	1.68 +/- 0.69 [%]
Orbital Inclination (inc)	81.07 +/- 1.64
Airmass coefficient 1 (a1)	0.963 +/- 0.011
Airmass coefficient 2 (a2)	0.029 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.1 %
Best Comparison Star	#2 - [156, 240]
Optimal Aperture	3.2
Optimal Annulus	12.68
Transit Duration (day)	0.0441 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.027 vs 0.1594 (deviation 1.06σ)
Duration fit vs published	0.0441 d vs 0.0512 d (deviation 0.87σ)
Mid-time O–C	0.0 min
Depth SNR	4.7
Residual scatter	1.1 %

Light curve

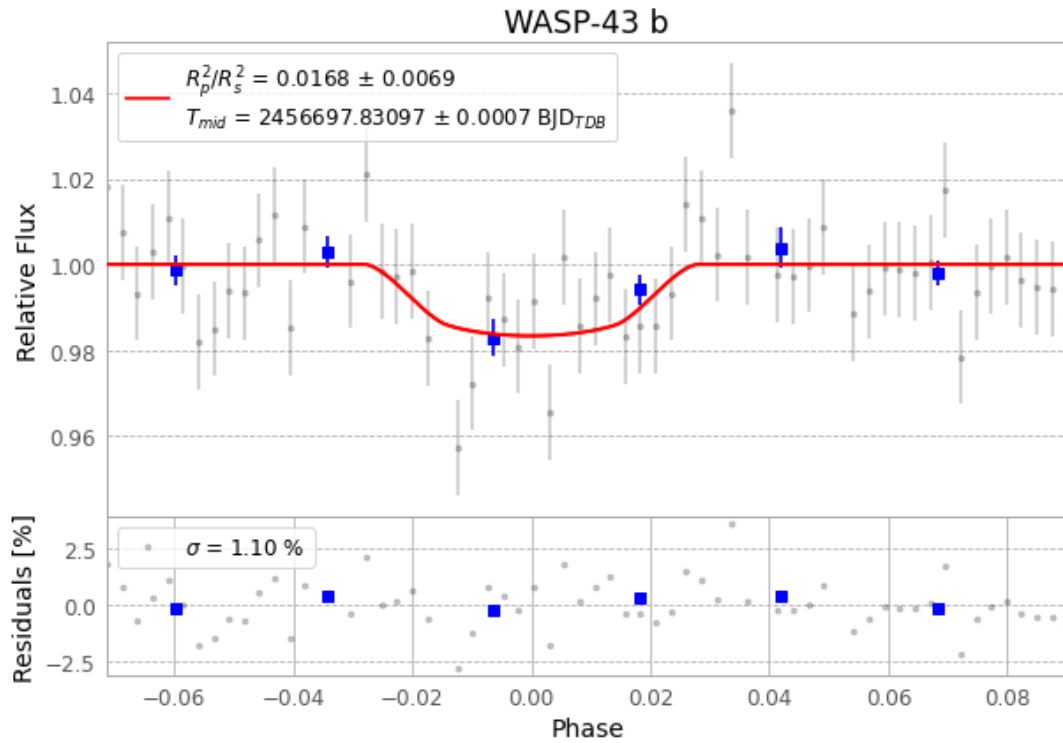


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-08.png (SHA-256 ace6142a18034668...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [222, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d7e3ee2dadee329...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), WASP-43140209062412.FITS → WASP-43140209093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140209.log (d673eec3b030...), FinalLightCurve_WASP-43 b_2014-02-08.png (ace6142a1803...), FinalLightCurve_WASP-43 b_2014-02-08.csv (fe4ccbc8601a...)

Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:52Z.